The Reconstructed Theory Has One Mass: a Machine-Checked Volume-Uniform Spectral Gap for a Concrete Transfer Operator

Lluís Eriksson*

DRAFT — not for distribution; live slots present

Abstract

For the spatial $\mathbb{Z}_2$ (Ising-slice) system in the Dobrushin window $2\tanh|\beta| + 2\tanh|\gamma| \le \alpha < 1$, we machine-check in Lean 4 that the Osterwalder–Schrader (site-form) reconstruction of the transfer operator carries *one* mass: there is a single $m > 0$ such that for *every* spatial extent L the reconstructed (site-form) transfer operator is unitarily conjugate — by the explicit $\sqrt{w}$ boundary dressing — to the symmetrised kernel whose projected norm the Dobrushin corollary bounds by e^{-m}, and every connected correlator of the transported transfer operator decays as $|\mathrm{connCorr}\, T\, \Omega\, v\, n| \le \|v\|^2(e^{-m})^n$ — constant per-observable, rate alone uniform in the volume. All quantifiers live in the Lean statements; the prose here copies the kernel's cut. To situate the claim in the 2026 landscape of machine-checked constructive field theory: the abstract construction (reflection-positive data → physical Hilbert space → self-adjoint contractive transfer operator) has been machine-checked, sorry-free, in the Harvard formalization programme, *with the spectral gap carried as an undischarged hypothesis* (`GappedTransfer`); the Osterwalder–Schrader axioms have been machine-checked for the free field; and measure-side Dobrushin clustering has been machine-checked for lattice Yang–Mills. What we add, and what to our knowledge no machine-checked artefact yet contains, is the combination {concrete interacting model, gap *proved* rather than hypothesised, uniform in the spatial volume} on the reconstructed operator — i.e. a concrete discharge of exactly the hypothesis shape the abstract frameworks carry. We do not claim Wightman reconstruction, a continuum limit, $SU(N)$, or any progress on the Clay problem.

1 Introduction and landscape

[SLOT: 2–3 paragraphs: the reconstruction idea; why volume-uniformity is the content (a gap at every fixed extent is classical and already in the programme's paper 8; ONE m for all extents is what a thermodynamic limit can consume); the D-6 input.]

1.1 Machine-checked prior work, cited before a referee does

The following table is load-bearing and its rows were collected by a four-modality search on 2026-08-04 (evidence: `docs/OS-R-PRIORITY-COLLATION.md` in the repository).

*With machine-checked artefacts produced on the programme's Colab plane; toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib pinned to `0764272048...`. Repository: THE-ERIKSSON-PROGRAMME, branch `d3-closure`, anchor `8127e5b52d75116b19c172ea022e12c1cab0f8c2`.

Artefact	What is machine-checked	What is not
`mrdouglasny/reflection-positivity` (sorry-free ∼2026-06)	RP data $\to$ Hilbert space (completion/quotient) $\to$ self-adjoint contraction T; trace formulas; susceptibility bounds from a gap	the gap itself: carried as the `GappedTransfer` *hypothesis*; no spatial-volume-uniform statement; concrete instance is finite-lattice RP only
`OSforGFF` (arXiv:2603.15770)	all five GJ/OS axioms for the d=4 Gaussian free field, 0 sorries	reconstruction (listed as next milestone); any interacting model
`xiyin137/OSreconstruction`	Wightman-side GNS lane of the full OS↔Wightman theorem	the theorem: 53 sorries + 12 axioms at last README snapshot
`mrdouglasny/lgt` (complete 2026-05-04)	Chatterjee-style Dobrushin uniqueness for lattice YM, $U(n)$, with exponential clustering whose constants depend only on (n, d, β)	an operator gap; any reconstruction; the statement is per-volume on the measure side
`pphi2`	transfer-matrix spectral machinery for φ^4_2	rests on 27–29 project axioms
This paper	the $\sqrt{w}$-dressing unitary; the transported volume-uniform gap; volume-uniform clustering of the reconstructed operator (`os_reconstruction_uniform_gap`, `os_reconstruction_uniform_clustering`; core 8481, oracle 3017, 0 sorry)	Wightman axioms; continuum limit; $SU(N)$; Clay; the raw-measure n-step identification (named next step)

2 The two geometries and the forced operator

[SLOT: Cite the programme modules: SpatialReflection (paper 12), SpatialOS (paper 13), SpatialReconstruction (paper 14) — the site form weighs by $1/w$, the forced operator $T = D_w K$, self-adjoint at every β, PSD for $\beta \geq 0$. State the defining equation as the Lean theorem names. NO new mathematics in this section.]

3 The dressing unitary

All three statements are green on the plane (module `YangMills/OS/OSReconstructionUniform.lean`, sha256 `5445429f...`, five-pass ladder, `ELAB_EXIT 0`). Write $Qu := \sqrt{w}\,u$ for the boundary dressing of a real Euclidean vector (`qEmbed`); the site form weighs by $1/w$.

Theorem 3.1 (A1, `siteForm_qEmbed`). *For $w > 0$ and real u, v:* $\mathrm{siteForm}\,w\,(Qu)\,(Qv) = \sum_\sigma u_\sigma v_\sigma$. *Per site the cancellation is elementary:* $\sqrt{w}u \cdot \sqrt{w}v \cdot (1/w) = uv$.

Theorem 3.2 (A2, `transferOp_qEmbed`). $T(Qu) = Q\big(\mathrm{act}\,(\mathrm{symWeighted}\,w\,\beta)\,u\big)$ *pointwise: the forced transfer operator on a dressed vector is the dressing of the symmetrised kernel's action.*

Theorem 3.3 (A3, `transferOp_iterate_qEmbed`). $T^{[n]}(Qu) = Q\big((\mathrm{act}\, S)^{[n]}u\big)$ *— the shape the in-tree measure bridge* ***gibbsPathSum_eq_iterate*** *consumes.*

4 The transported volume-uniform gap

Theorem 4.1 (`os_reconstruction_uniform_gap`). *In the window* $2\tanh|\beta| + 2\tanh|\gamma| \le \alpha < 1$ *there is* $m > 0$ *such that for every* L *there are* $\lambda > 0$ *and* $\Omega > 0$ *with the fixed-point equation* $\sum_\tau \mathrm{tiltKernel}\,(\mathrm{sliceW}\,\gamma\,L)\,\beta\,\lambda\,\sigma\,\tau\,\Omega_\tau = \Omega_\sigma$*, the projected bound* $\|\mathrm{projectedTransfer}\ldots\| \le e^{-m}$*, and the A2 identity at* $w := \mathrm{sliceW}\,\gamma\,L$.

The proof is an `obtain` on `dobrushin_ising_uniform_gap` followed by A2: *the same witnesses*, no new analytic content — which is precisely the point. The uniformity is real because it is D-6's; the reconstruction contributes the identification, not a new estimate.

5 Identification against the measure

Green at pass 5 (operator side):

Theorem 5.1 (B0b, `act_symWeighted_eq_smul_act_tilt`). $\mathrm{act}\, S\, u = \lambda \cdot \mathrm{act}\,(\mathrm{tilt})\, u$ *for* $\lambda \neq 0$.

Theorem 5.2 (B0a, `act_iterate_eq_opOf_pow`). $(\mathrm{act}\, M)^{[n]}u = (\mathrm{opOf}\, M)^n(\mathrm{toLp}\, u)$ *pointwise.*

Theorem 5.3 (`os_reconstruction_uniform_clustering`). *In the window: ONE* $m > 0$ *such that for EVERY spatial extent* L*, every connected correlator of the transported transfer operator obeys* $|\mathrm{connCorr}\, T\,\Omega\, v\, n| \le \|v\|^2 \cdot (e^{-m})^n$*. The constant is per-observable; the rate alone is uniform in* L*.*

The packaging is by citation: `vacuumTransfer_opOf` + `tiltKernel_symm` + `clustering_of_gap`, all in-tree. **[SLOT: The gibbsPathSum-level identification (raw Gibbs two-point sums against the operator powers through B0a/B0b and A3) is the one named remaining step of the charter's part B; state here exactly what is proved when it lands, and nothing more.]**

6 Reproduction

All numbers below were read from committed run artefacts (ledger Addenda 616–618); predictions were committed *before* their measurements.

Quantity	Predicted	Measured
core jobs, unwired	8480 (ledger baseline)	8480
core jobs, wired	$8480 + 1$	8481
oracle reports, wired	$3010 + 7$	3017
`sorryAx` occurrences	0	0
OS-R endpoint axiom sets	standard triple	exactly `[propext, Classical.choice, Quot.sound]` $\times 7$

Module: `YangMills/OS/OSReconstructionUniform.lean`, sha256 `5445429f7736...30cad1` (12,347 bytes), five elaboration passes on a fresh high-RAM Colab runtime (toolchain `v4.29.0-rc6`, Mathlib pinned); wired at anchor `8127e5b52d75116b19c172ea022e12c1cab0f8c2`. The count prediction $+1/+7$ held exactly — the ninth consecutive exact count prediction of the programme's D/OS lane. **[SLOT: Fresh-clone second witness at the anchor: judges both modes, core total, oracle report count with a continuation-joining axiom checker, and the sha256 list — from the pass-7 run output.]**

7 What is not claimed

No Wightman axioms, no OS→Wightman analytic continuation, no continuum limit, no $SU(N)$, no statement about the Clay problem. The window is the Dobrushin window; outside it the programme's own kill-test (repository, Addendum 610) shows the standard-cone Birkhoff route is projectively blind to the site weight, which is why no claim beyond the window appears here.